Lesson plan · for the teacher

Big problems, good answers

Optimization · Level 4 · Lesson 4

Time
40 min

Materials
Board and chalk; A name card for each pupil in one group (about 12); A seating grid drawn on the board

Learning goals

- Improve a plan in small steps, and see why it can get stuck.
- Escape a stuck plan by allowing some worse moves.
- Decide when good enough is enough, and plan a project with people.

Activity: Swap to improve

1. Warm-up (5 min): Ask pupils to guess how many ways 12 pupils can sit at 12 desks. Reveal 479,001,600, and explain why nobody checks them all.
2. Set up (7 min): Draw 12 desks in 3 rows of 4. Agree pairs who work together (each pair sitting apart costs the number of steps between their desks) and chatty pairs (side by side costs 4).
3. Climb (10 min): Groups place name cards at random, work out the cost, then swap two cards at a time, keeping a swap only if the cost goes down. Stop when no swap helps and write the cost.
4. Escape (10 min): Now groups may keep up to three swaps that make the cost a little worse, early on only, then go back to better-only swaps. Write the new cost.
5. Wrap-up (8 min): Compare costs across groups. Who got stuck? Did the worse swaps help? Agree what cost would be good enough and why.

Prepare the pairs in advance on a card (who works together, who is chatty) so groups score plans the same way. Use made-up names if pupils might be upset by who sits where.

Discussion

- Why can a plan be stuck even though no swap improves it?
- Who should decide when our timetable is good enough?
- What rules might our data not know about?

Exit ticket

- Write the first two steps you would take to lead an optimization project at our school.

Try it in class tomorrow
Draw a seating plan on the board with pupils' names on cards, and agree a simple cost (for example, one point for every pair of friends who work together sitting apart). Pupils swap two cards at a time and keep a swap only if the cost goes down. When they're stuck, allow one worse swap and see whether they find a better plan.

Worksheet · for pupils

Big problems, good answers

Optimization · Level 4 · Lesson 4

Name _____ Class _____ Date _____

1 **The solver for next week's exam invigilation rota has run for an hour. Its best rota is within 2% of a bound no rota can beat, and the rota is due tomorrow morning. Do you keep it running?**

- ◯ Yes: run until it proves the best
- ◯ No: 2% is good enough here; check it with people
- ◯ Start again from scratch

2 **Put the steps of a school optimization project in order.**

- ☐ Pilot it, explain it and decide together
- ☐ Test the model on last year's plan
- ☐ Collect the data and build a small model
- ☐ Agree the goal and the fairness rules, before any plan is seen
- ☐ Agree the problem with the people it affects

3 **Complete the simulated annealing rule: always keep a swap that isn't worse, sometimes keep a worse one, and cool down slowly.**

```
d = new_cost - cost
if d <= _____ or random.random() < math._____(-d / T):
    plan, cost = new_plan, new_cost
T = T * _____
```

Word bank: `0` `1.5` `exp` `log` `0.999` `1`

4 **If hill climbing stops and no single swap improves the plan, the plan is the best possible.**

- ◯ True
- ◯ False

5 **You are leading an optimization of next term's duty rota. Which of these should you decide before you start the solver? Choose all that apply.**

- ☐ The goal and the fairness rules, agreed with staff
- ☐ How long the solver may run, and what gap is good enough
- ☐ Who makes the final call, and how to appeal
- ☐ Which teacher should get the worst duty
- ☐ To hide the rules so nobody games them

✎ **Which plan at your school is too big to check by hand: the timetable, invigilation, the bus? What would good enough mean for it, and who should decide that?**